FIXING YOUR ENDER 3 V2

An instruction manual for humans.

One action per step. Follow it in order.

INSIDE

PHASE 1	**Is the printer crooked?**	10 min · free · do this first
PHASE 2	**Level the bed**	15 min · the part everyone does badly
PHASE 3	**Fix the extrusion**	15 min · name the symptom first
PHASE 4	**If it was moisture**	4–6 hrs of waiting
PHASE 5	**If 1 and 2 failed**	the buy-a-part stage

BEFORE YOU START

TIME
About 45 minutes.

DIFFICULTY
Moderate. The order is what matters.

PAGES YOU NEED
3–7 for the bed. 8–10 only if it also extrudes badly.

TOOLS
- ☐ Ruler or tape measure
- ☐ Two identical objects — two mugs, two cans, two blocks
- ☐ One sheet of ordinary printer paper
- ☐ The hex key and wrench that came with the printer
- ☐ A marker or pen
- ☐ Phone (for photos)

FOUR RULES

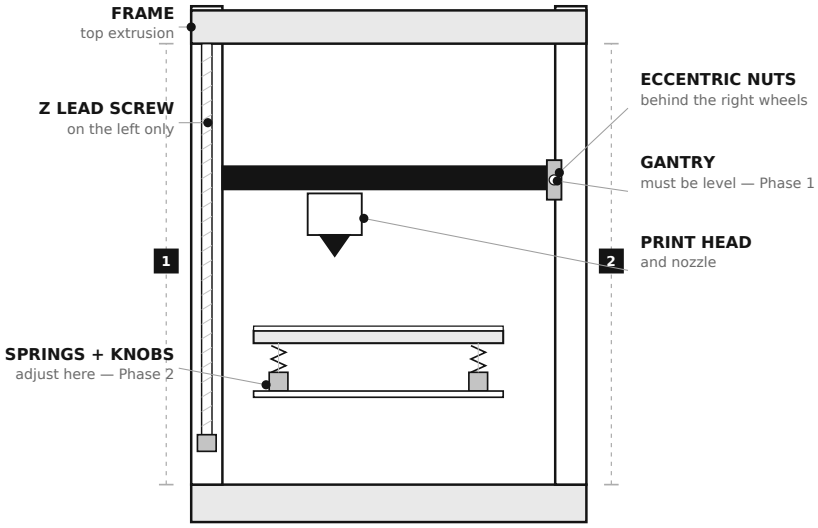
01 **Do the phases in order.** Phase 2 is impossible if Phase 1 is wrong.

02 **One change at a time.** Change two things and you won't know which one worked.

03 **Write the number down before you change it.** Every time.

04 **When a step says STOP, stop.** That's where the money gets wasted.

> **■ THIS IS A REAL FIX, SO LOG IT**
> When you finish, fill in page 12. What was actually wrong, what fixed it, and what you tried that didn't work. Then photograph the first good print. That's the proof — and it's worth as much as the repair.



Front view, schematic, not to scale. Dashed lines ① and ② are the measurements in steps 1.4 and 1.5.

PHASE 1

Is the printer crooked?

1

10 minutes · free · do this first

If the arm that holds the nozzle is tilted, one side of the bed will never be reachable. You'd be adjusting the bed against a crooked ruler. This is the most common reason a bed "can't be leveled" — and it costs nothing to check.

1.1 Turn the printer **off**. Wait 5 seconds.

1.2 Turn it back on. On the screen: **Prepare → Disable steppers.** The motors now let you move things by hand.

1.3 Stand facing the printer. Notice the **horizontal bar** across the top that the nozzle hangs from. That's the gantry. You're checking that it's level.

1.4 Hold the ruler upright against the **left** vertical post. Measure from the **base** up to the **bottom edge of the gantry bar**.

LEFT = mm

1.5 Do the exact same thing on the **right** side.

RIGHT = mm

1.6 Compare the two numbers.

DECISION POINT

If LEFT and RIGHT are within 1 mm of each other:

✓ Your gantry is fine. **Skip to Phase 2.**

If they differ by more than 1 mm:

✗ Your gantry is tilted. That is the actual cause of your bed problem. **Continue to step 1.7.**

1.7 Find the **two eccentric nuts** on the right side of the gantry — the odd nuts behind the wheels that ride the right post. They have a flat side and turn with the wrench to loosen or tighten the wheel grip.

1.8 Loosen both eccentric nuts a **quarter turn**. The right side of the gantry should now move slightly.

1.9 Grip the **right end** of the gantry bar with one hand and move it **up or down** until the right measurement matches the left.

> **■ GRAVITY DOES THIS**
>
> Whichever side is lower is the one that sagged. It isn't your fault and it isn't damage.

1.10 Hold it in place. Tighten both eccentric nuts until the wheels grip firmly — the gantry shouldn't wobble, but the wheels should still turn.

1.11 Measure left and right again.

DECISION POINT

Within 1 mm? ✓ Phase 1 complete — go to Phase 2.

No? Repeat 1.8–1.11 once. Still no? Stop and photograph the wheel assembly — something else is bent. See Phase 5.

PHASE 2

Level the bed

2

15 minutes · the part everyone does badly

Most people adjust one corner, then the opposite corner, and undo their own work. Use this order and **do the whole sequence twice**. The first pass always throws the others off. That's normal, not a mistake.

2.1 Turn the four knobs under the bed until the springs are about **half squashed** — roughly 10 mm of spring showing. Do all four by eye. Don't level yet.

2.2 On the screen: **Prepare → Auto Home.** Wait for it to finish.

2.3 **Prepare → Disable steppers** again.

2.4 Slide the print head by hand so the nozzle sits over the **front-left** corner, about 2 cm in from the edge.

2.5 Slide the paper under the nozzle.

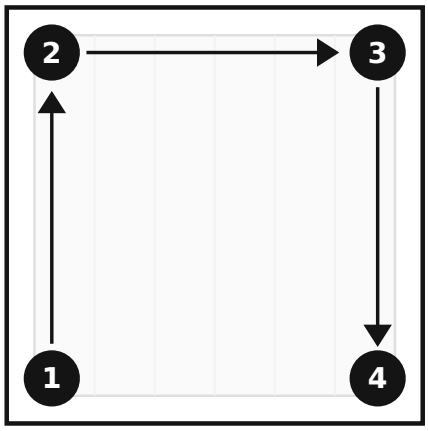
2.6 Turn the knob under that corner **slowly** until the paper slides out with a **slight drag** — like a fingernail on a table. Not gripped, not loose.

2.7 Move to the **back-left** corner. Repeat 2.5 and 2.6.

2.8 Move to the **back-right** corner.

2.9 Move to the **front-right** corner.

2.10 **SECOND PASS — now do 2.4 through 2.9 again.** The first pass always throws the others off. This is normal, not a mistake.



FRONT OF THE PRINTER

1 → 2 → 3 → 4
clockwise from front-left

PASS 1
rough the corners in

PASS 2
recheck all four — this is the real one

Four corners, clockwise from the front left. Two passes, then leave it alone.

■ WHY TWICE

Raising one corner bends the plate against the other three, so the corner you set first is always slightly off by the time you reach the fourth. The second pass is where the bed actually gets flat — skip it and you'll be back here in a week.

DECISION POINT

All four corners drag the paper, and no knob is loose or wobbly:

✓ Bed is level. **Go to Phase 3.**

One knob has to be so loose it's falling off — or is already loose:

✗ **STOP.** Do not keep turning. That is your original complaint, and it means one of the following is true.

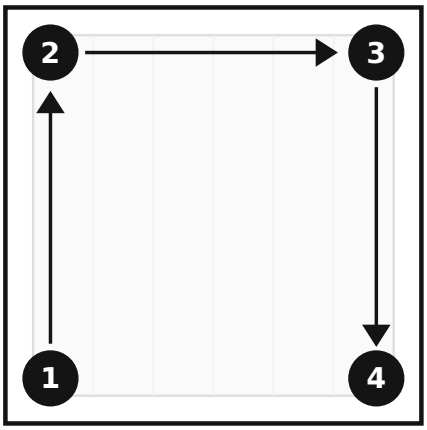
Check	How	Fix
Screw head not seated	Look at the top of the bed. All four heads sunk flat?	Push it down; check it's the same length as the rest
Spring missing or wrong	All four springs the same length?	Match them, or replace all four
Knob threads stripped	Does the loose knob spin without gripping?	Replace the knob
Bed plate warped	Straightedge across the bed, corner to corner	Silicone spacers, or an ABL probe
Y carriage bent	Remove the bed; straightedge across the plate underneath	Replace the carriage plate

> **■ CHECK PHASE 1 AGAIN BEFORE BLAMING A BENT PART**
>
> A tilted gantry mimics a bent carriage almost perfectly, and it's free to fix. Most people buy parts for a problem that was alignment all along.

KEEP THIS BY THE PRINTER

THE PAPER TEST

Four corners. Clockwise from the front left. Then the whole loop again.



FRONT OF THE PRINTER

1 → 2 → 3 → 4
clockwise from front-left

PASS 1
rough the corners in

PASS 2
recheck all four — this is the real one

Paper does this	Means	Turn the knob
Slides out with a slight drag	Correct	Nothing — next corner
No contact at all	Nozzle too high	Tighten the spring (raise the bed)
Pinned, can't move	Nozzle too low	Loosen the spring (lower the bed)
Right at one corner, wrong at the opposite	Gantry tilt	Phase 1 — don't touch the knobs
Perfect, then wrong again in ten minutes	Weak springs or stripped knob	Phase 5 — silicone spacers

Chasing a perfect paper feel on pass five is how people spend an hour and end up worse than pass two. If it's even and it drags, print the first-layer test. The print tells the truth; the paper is only a rough guide.

PHASE 3

Fix the extrusion

3

15 minutes · name the symptom first

Nobody can fix "extrusion problems." Name the symptom first. One test splits it in two.

3.1 Load a **brand-new, unopened** spool of filament.

3.2 Print a small test cube or a benchy.

3.3 Look at the result.

DECISION POINT

It prints clean now → it was **moisture**. Go to **Phase 4**, then come back and finish.

It still prints badly → it's **mechanical**. Moisture was a red herring. **Continue to step 3.4.**

3.4 Look at the nozzle. Is plastic oozing from around the **outside** of it, or curling out sideways instead of straight down? If yes, the tube isn't sealed against the nozzle — do 3.5 to 3.8.

3.5 Heat the nozzle to printing temperature: **Prepare → Preheat → PLA**.

3.6 Loosen the fitting that holds the white tube, by **half a turn**.

3.7 Push the tube **firmly down** so it presses flat against the back of the nozzle. Hold it there.

3.8 Tighten the fitting back up while still pushing down. Done.

STILL BAD? DO A COLD PULL.

01 Heat to 200 °C

02 Push filament through by hand until it extrudes

03 Set the temperature to 90 °C and wait

04 Pull the filament out firmly, in one motion

05 Look at the tip — it should have a clean point with the gunk stuck to it

06 Cut that end off, reload, test print

Repeat twice if the first pull comes out black.

3.9 Still bad? **Check the extruder arm** — the plastic lever on the motor that grips the filament. Remove it and look at the **underside** for a hairline crack. It cracks where you can't see it, and it fails exactly like a clog.

3.10 Still bad? Check the brass gear on the motor for flattened or shiny worn teeth. Replace if worn.

PHASE 4

If it was moisture

4

4–6 hours of waiting · 5 minutes of work

4.1 Put the spool in a food dehydrator or a filament dryer.

4.2 Set it to **40–45 °C.**

> **▲ DO NOT GO ABOVE 50 °C**
>
> PLA starts going soft around 60 °C. A dehydrator set to "jerky" will warp the spool and ruin the filament. Low and slow.

4.3 Leave it **4 to 6 hours**.

4.4 Move it straight into a sealed container with desiccant packets. It reabsorbs moisture within hours if left out.

4.5 Add a cheap hygrometer inside the container. Under 20% is good.

> **■ DESICCANT DOES NOT DRY FILAMENT**
>
> It only keeps dried filament dry. You need heat **and** a sealed box. Skip the heat and nothing improves; skip the box and it comes back by the weekend.

PHASE 5

If phases 1 and 2 failed

5

The buy-a-part stage. Only now.

5.1 Re-check Phase 1. Be honest about the measurement.

5.2 Remove the bed. Lay a straightedge across the metal plate underneath, in both directions. Bent means replace.

5.3 Lay the straightedge across the bed plate itself, corner to corner. A dish in the **middle** is normal. A raised or sunken **corner** is a warp.

5.4 Order accordingly:

Part	~Price	Fixes
Silicone bed spacers + nyloc nuts	$10–15	Springs that hold tension; knobs that stop walking loose
Metal extruder arm	$15	Cracks in the stock plastic arm
Spare nozzles (pack)	$5	Clogs
Bed carriage plate	$15	Bent Y carriage
ABL probe (CR Touch)	$35	A warped bed surface

> **■ AN ABL PROBE WILL NOT FIX YOUR PROBLEM**
>
> It does **not** fix a tilted gantry and it does **not** fix a corner that can't hold tension. It only compensates for a warped surface. Buying one to solve a leveling problem is the most common way people waste $35 and three weekends.

THE BUDGET LADDER — SPEND IN THIS ORDER

Spend	Gets you	Stop here if...
$0	Gantry level, bed leveled twice, tube seated, cold pull, new spool test	It prints. Most machines are fixed at $0.
$10–15	Silicone spacers + nyloc nuts	The bed holds. It usually will.
$15	Metal extruder arm	Your plastic arm is cracked.
$35	ABL probe	You have confirmed a **warped surface** and nothing else.

WHEN TO STOP AND TAKE IT SOMEWHERE

There is a point where continuing costs more than the machine. Knowing the number is what separates a repair from a project. These three symptoms mean it's not you:

Symptom	What it usually is	Realistic call
Z banding on every print, or layers shifting	Bent Z screw, or a failing stepper driver	Board-level — worth a shop, not a weekend
Frame out of square even with the gantry level	Dropped or over-tightened during assembly	Difficult but fixable — loosen everything and square it
Bed surface warped more than about 0.3 mm	Trammed from the factory	Glass or a new plate — not adjustable

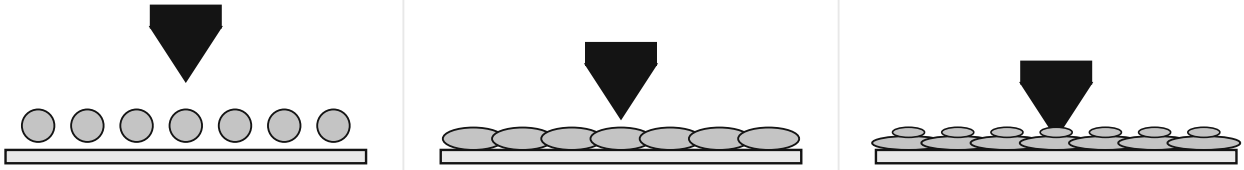
A shop bench fee runs $80–120, and a new Ender 3 V2 is $170–200. Once you're past the $35 rung, check that number before you spend a fourth weekend. Not every machine is worth saving — and saying so out loud is the same professional judgment you'd use on anyone else's printer.

FINAL CHECK

1 · TOO HIGH
gaps between the lines

2 · CORRECT
squashed, lines touching

3 · TOO LOW
ridged, rippled, see-through



Cross-section through the first layer. This is the only test that tells the truth.

Print something with a large flat first layer — a one-layer square test, or a benchy.

What a good first layer looks like: a smooth, slightly squashed ribbon. Lines touching, no gaps, no ridges you can catch a fingernail on.

Looks like	Meaning	Do this
Gaps between the lines	Nozzle too high	Raise the bed slightly at that corner
Rough, rippled, transparent	Nozzle too low	Lower the bed slightly
Good at the front, bad at the back	Bed isn't flat	Phase 2, second pass
Bad on one side only	Gantry still tilted	Phase 1

WHAT TO PHOTOGRAPH

The log entry is only worth what you can show. Four shots, taken with the phone you already have:

☐ The problem, before you touch it — the loose knob, the stringy print, the error on screen
☐ The part that turned out to be at fault, out of the machine and in your hand
☐ The fix, halfway through
☐ The first good print, straight on

Four photos and five minutes of writing turn a fix you did for yourself into the first line of a work history.

WHEN YOU'RE DONE

ENTRY 001 · ENDER 3 V2 · BED LEVELING AND EXTRUSION

Write this down. It takes five minutes and it's the part that turns a repair into proof that you can finish a difficult thing.

Matching entry in the repository: fix-log/001-ender-3-v2-bed-and-extrusion.md

What was actually wrong (root cause)

What fixed it

Time spent / parts cost / what I'd do differently

What I tried that didn't work (dead ends)

> **■ THEN PHOTOGRAPH IT, THEN TYPE IT UP**
>
> The before, the after, the first good print. That's a fix with a documented cause and a documented result. Copy the same answers into fix-log/001-ender-3-v2-bed-and-extrusion.md and mark it solved — that file is the first line of a work history.

QUICK REFERENCE

START HERE IF YOU DON'T KNOW WHY

Symptom you reported	What it means	Phase
Bed won't level; one screw has to be so loose it falls off	That corner is maxed out — alignment, not adjustment	1, then 2
Bed won't stay level after adjusting	Weak stock springs, or a stripped knob	5
Extrusion problems, suspected humidity	Moisture — or five mechanical causes that look identical	3, then 4
Looks fine, prints badly	Bed surface warp, or gantry tilt	1, then 5

BY SYMPTOM

Situation	Go to
Bed won't level, one corner maxed out	**Phase 1** — almost always the gantry
Bed won't stay level	Phase 5 — silicone spacers
Under-extruding, clicking, gaps	Phase 3
Popping, crackling, rough surface	Phase 4 — moisture
Poor layer quality, one side worse	Phase 1
Tried everything	Phase 5
Don't know	Step 3.1 — a new spool splits it in two

Northstar Fix Log · Entry 001 · Ender 3 V2
Source: fix-log/manuals/ender-3-v2-manual.md